MILITARY SPECIFICATION

ADAPTION KIT SUBASSEMBLY

CONTROL UNIT, M4

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of control unit entitled Adaption Kit Subassembly: Control Unit, M4 which is used in the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-N-411	- Nitrogen, Technical.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDNANCE CORPS

8829845	- Cover, Control Unit.
8830142	- Adaption Kit Subassembly: Control Unit, M4.
8838334	- Control Housing Assembly.
EL-8854157	- Equipment List.

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The control unit shall comply with all requirements specified on the drawing (dwg) 8830142, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SMUPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Continuity and resistance. The continuity and resistance between terminals of the control unit shall not exceed the resistance specified below when tested as specified in 4.4.1.

From Terminal	To Terminal	Max Resistance (Ohms)
2J1-B	2J3-v	0.15
2J1-C	2J1-D	0.15

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From Terminal	To Terminal	Max Resistance (Ohms)
2J1-E	2J4-n	0.15
2J1-j	2J3-f	0.15
2J1-G	2J4-E	0.15
2J1-G	2J4-f	0.75
2J1-G	2J4-r	0.15
2J1-H	2J3-z	0.15
2J1-K	2J3-x	0.15
2J1-L	2J1-M	0.15
2J1-L	2J3-FF	0.15
2J1-R	2J4-K	0.15
2J1-U	2J3-M	0.15
2J1-V	2J3-K	0.15
2J1-e	2J3-b	0.15
2J1-j	2J4-n	0.15
2J1-k	2J1-n	0.15
2J1-K	2J6-c	0.30
2J1-m	2J2-A	0.75
2J1-n	Case	0.15
2J1-p	2J8-A	0.75
2J1-p	Case	0.15
2J1-q	Case	0.15
2J1-r	2J3-c	0.15
2J1-s	Case	0.15
2J3-A	2J5-q	0.15
*2J3-B	2J4-C	0.75
*2J3-C	2J4-J	0.75
*2J3-D	2J4-R	0.75
2J3-E	2J7-F	0.15
2J3-F	2J7-U	0.15
2J3-G	2J5-b	0.15
2J3-H	2J7-S	0.15
2J3-J	2J3-W	0.15
2J3-L	2J3-BB	0.30
2J3-N	2J6-d	0.30
2J3-R	2J4-i	0.15
2J3-T	2J3-j	0.15
2J3-U	2J3-a	0.30
2J3-U	2J6-Z	0.30
2J3-U	2J7-D	0.15
2J3-U	2J7-E	0.30
2J3-V	2J3-h	0.30
2J3-V	2J3-r	0.15
2J3-Y	2J5-F	0.15
2J3-Y	2J5-Y	0.15

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2J3-[illegible]	Case	0.15
2J3-d	2J7-G	0.15
2J3-e	2J7-R	2.00
2J3-h	2J7-T	0.30
2J3-i	2J6-T	0.15
2J3-i	2J6-U	0.15
2J3-j	2J3-h	0.30
2J3-m	2J3-t	0.15
2J3-n	2J6-C	0.15
2J3-n	2J6-D	0.15
2J3-p	2J7-W	0.30
*2J3-q	2J4-a	0.30
2J3-s	2J5-C	0.15
2J3-s	2J5-D	0.15
2J3-u	2J5-X	0.15
2J3-u	2J6-X	0.15
2J3-w	2J3-GG	0.15
2J3-y	2J3-DD	0.30
2J3-AA	2J3-EE	0.15
2J3-CC	2J7-B	0.30
2J3-CC	2J7-C	0.30
2J3-HH	2J5-T	0.15
2J3-HH	2J5-U	0.15
2J4-A	2J5-S	0.15
2J4-B	2J5-W	0.15
2J4-G	2J5-G	0.15
2J4-G	2J6-G	0.15
2J4-H	2J5-V	0.15
2J4-L	2J5-H	0.15
2J4-L	2J6-H	0.15
2J4-M	2J6-V	0.15
2J4-N	2J6-W	0.15
2J4-S	2J7-L	0.15
2J4-T	2J7-N	0.15
2J4-U	2J7-Z	0.15
2J4-V	2J7-V	0.15
2J4-X	2J7-P	0.15
2J4-Y	2J7-K	0.15
2J4-Z	2J7-M	0.15
2J4-b	2J7-b	0.15
2J4-c	2J7-c	0.15
*2J4-e	2J6-f	0.75
2J4-j	2J7-H	0.15
2J4-j	2J7-J	0.15
2J4-m	2J5-N	0.15
2J4-m	2J6-N	0.15
2J5-B	**2J6-B**	**0.30**
2J5-E	**2J5-a**	**0.15**
2J5-J	**2J6-J**	**0.15**
2J5-K	**2J6-K**	**0.15**

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2J5-L	2J6-L	0.15
2J5-R	2J6-S	0.15
2J5-Z	2J6-F	0.15
2J5-Z	2J6-Y	0.15
2J5-c	2J6-b	0.15
2J5-$\overline{d}$	2J3-$\overline{S}$	0.30
2J5-$\overline{e}$	2J3-P	0.15
2J5-$\overline{f}$	2J6-$\underline{g}$	0.15
2J6-$\overline{E}$	2J6-$\overline{a}$	0.15
2J6-R	2J7-$\overline{A}$	0.15
2J6-Z	2J3-a	0.30
2J6-$\underline{c}$	2J3-$\overline{g}$	0.15

*Initiator relays energized.

From Terminal	To Terminal	Resistance (ohms)	Test Code Nos.
2J1-G	2J4-f	0.75 max	030001
2J1-k	2J6-$\overline{c}$	0.30 max	030002
2J1-$\overline{m}$	2J2-$\overline{A}$	0.75 max	030003
2J1-$\overline{m}$	Case	0.15 max	030004
2J1-$\overline{p}$	2J8-A	0.75 max	030005
2J1-$\underline{p}$	Case	0.15 max	030006
2J3-$\overline{B}$	2J4-C	0.75 max	030007
2J3-C	2J4-J	0.75 max	030008
2J3-D	2J4-R	0.75 max	030009
2J3-L	2J3-BB	0.30 max	030010
2J3-N	2J6-d	0.30 max	030011
2J3-U	2J3-$\overline{a}$	0.30 max	030012
2J3-U	2J6-$\overline{Z}$	0.30 max	030013
2J3-U	2J7-E	0.30 max	030014
2J3-V	2J3-h	0.30 max	030015
2J3-e	2J7-$\overline{R}$	2.00 max	030016
2J3-$\overline{h}$	2J7-T	0.30 max	030017
2J3-$\overline{j}$	2J3-h	0.30 max	030018
2J3-$\underline{p}$	2J7-$\overline{W}$	0.30 max	030019
2J3-$\underline{q}$	2J4-a	0.30 max	030020
2J3-$\underline{x}$	2J3-$\overline{DD}$	0.30 max	030021
2J3-CC	2J7-B	0.30 max	030022
2J3-CC	2J7-C	0.30 max	030023
2J4-e	2J6-f	0.75 max	030024
2J4-$\overline{j}$	2J7-$\overline{H}$	0.15 max	030025
2J4-$\overline{j}$	2J7-J	0.15 max	030026
2J5-$\overline{B}$	2J6-B	0.30 max	030027
2J5-d	2J3-S	0.30 max	030028
2J6-$\overline{Z}$	2J3-$\underline{a}$	0.30 max	030029

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3.6.1 Circuit resistance. The circuit resistance of the control unit shall be within the limits specified below when tested as specified in 4.4.1.1.

From Terminal	To Terminal	Resistance (ohms)	Test Code Nos
2J1-P	2J4-L	6960 to 7040	031001
2J1-S	2J4-L	4972 to 5028	031002
2J1-k	2J3-a	5.7 to 6.8	031003
2J1-$\overline{K}$	2J3-$\overline{t}$	5.5 to 6.6	031004
2J1-$\overline{k}$	2J3-$\overline{u}$	5.5 to 6.6	031005
2J1-$\overline{E}$	2J3-$\overline{y}$	5.5 to 6.6	031006
2J3-$\overline{L}$	2J6-$\overline{f}$	1.3 to 1.9	031007
2J3-d	2J3-$\overline{w}$	5.5 to 6.6	031008
2J3-$\overline{n}$	2J3-$\overline{EE}$	5.5 to 6.6	031009
2J3-$\overline{n}$	2J7-H	5.7 to 6.8	031010
2J1-$\overline{K}$	2J3-K	11.0 to 13.0	031011
2J1-$\overline{K}$	2J3-k	11.0 to 13.0	031012
2J4-$\overline{L}$	2J5-$\overline{J}$	2 K $\pm$ 0.2%	031013
2J4-L	2J5-K	4 K $\pm$ 0.2%	031014
2J4-L	2J5-L	6 K $\pm$ 0.2%	031015
*2J4-W	2J4-d	5600 min	031016
2J6-f	2J7-$\overline{T}$	310 $\pm$ 5%	031017

*Not to be checked at temperature extremes.

3.7 Diode characteristics. Forward voltage drop across the diodes and the reverse voltage drop across a 150 $\pm$ 1.5 ohm resistor shall be as specified below when tested as specified in 4.4.2.

Connect Positive Lead To	Connect Negative Lead To	Applied Forward Current	Maximum Forward Volt Drop	Applied Reverse Volt	Max Volt Drop Across Series Resistor
2J3-j	2J3-d	10 $\pm$ 0.5 Amp	2.25V		
2J3-r	2J3-d	10 $\pm$ 0.5 Amp	2.25V		
2J3-d	2J3-j			100 $\pm$ 5V	1.5V
2J3-d	2J3-r			100 $\pm$ 5V	1.5V

Positive Lead to Terminal	Negative Lead to Terminal	Applied Forward Current	Maximum Forward Voltage Drop	Test Code Nos
2J3-j	2J3-d	10 $\pm$ 0.5 A	2.25	040001
2J3-r	2J3-d	10 $\pm$ 0.5 A	2.25	040002

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Positive Lead to Terminal	Negative Lead to Terminal	Applied Reverse Voltage	Max Forward Drop Across Series Res	Test Code Nos
2J3-d	2J3-j	100 ± 5	1.5	040003
2J3-$\bar{d}$	2J3-$\bar{r}$	100 ± 5	1.5	040004

3.8 Insulation resistance. The insulation resistance of the control unit shall be greater than 20 megohms between each terminal listed below and all other terminals listed and the control housing when tested as specified in 4.4.3.

Receptacle 2J1 - B,C,E,H,L,P,R,U,V,$\bar{e}$,$\bar{j}$,$\bar{n}$,$\bar{r}$
Receptacle 2J3 - A,E,F,G,H,W,Y,$\bar{i}$,$\bar{n}$,$\bar{s}$,$\bar{CC}$,$\bar{HH}$,P,R,$\bar{e}$
Receptacle 2J4 - A,B,C,D,F,G,H,$\bar{J}$,$\bar{M}$,$\bar{N}$,R,S,T,U,V,$\bar{X}$,Z,$\bar{a}$,$\bar{b}$,$\bar{c}$,$\bar{d}$,$\bar{e}$, $\bar{m}$,$\bar{r}$,Y
Receptacle 2J5 - $\bar{B}$,$\bar{E}$,R,Z,$\bar{c}$,$\bar{d}$,$\bar{f}$
Receptacle 2J6 - E,R,$\bar{d}$,$\bar{e}$

Unused pins
Receptacle 2J1 - A,F,J,N,T,W,X,Y,Z,$\bar{a}$,$\bar{b}$,$\bar{c}$,$\bar{d}$,$\bar{f}$,$\bar{g}$,$\bar{h}$,$\bar{i}$ and $\bar{t}$.
Receptacle 2J4 - P,$\bar{k}$,$\bar{q}$,$\bar{g}$, and $\bar{h}$.
Receptacle 2J5 and 2J6 - A,M, and P.
Receptacle 2J7 - X, Y, and $\bar{a}$.

Unused terminals on any one connector shall be connected together and tested as one terminal.
Connector shall be a ground potential.

3.9 Pressure retention. The control unit shall withstand a pressure of 15 ± 1.5 pounds per square inch gage (psig) without a pressure loss exceeding two psi per hour when tested as specified in 4.4.4.

3.10 Operation. With the baroswitch, tdd, warhead and parameter insertion simulators shown in Figure 1 through 4 connected to the control unit, and with the switches closed as specified in Table I, the control unit shall provide to the warhead simulator the voltages as specified in Events 9, 10 and 12. The testing shall be as specified in 4.4.5.

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Table I

Control Function	Time (Seconds)	Simulator Function	Function Measurement	Monitor Terminals
Initiation capaitor charge	X-120 ± 10	Close S7		
Disconnect power	X = 0	Open S7		
Timer initiation and initiation safing	X + 3 ± 1	Close S8	Open continuity loop	2J3-e to 2J7-$\bar{R}$
TDD filament battery detonate	T3		Detonate pulse	2J5-d to 2J5-$\bar{f}$ 2J6-$\bar{d}$ to 2J6-$\bar{f}$
Safing function	T3 + 34 ± 7	Close S3		
WHS battery initiation	T3 + 58 ± 8		24.0 volts minimum 32.0 volts maximum 30 seconds after voltage rise	2J3-r to 2J3-$\bar{h}$ 2J3-$\bar{j}$ to 2J3-$\bar{h}$
Fuze safing	During battery voltage rise		Continuity at time of closure	2J3-DD to 2J7-C
Arming function	X + 175 ± 5	Close S2		
A-1 and B1	T8 + 0.1 ± 0.05		6vdc min for 40 ms min	2J1-L to 2J1-k
A2 and B2			28 ± 4vdc steady state 10V min during A3, B3 surge but never less than A3, B3 voltage	2J1-M to 2J1-k 2J1-$\bar{U}$ to 2J1-k 2J1-$\bar{V}$ to 2J1-k
TDD plate battery detonate			Blown fuzes	2J5-Z to 2J5-c 2J6-$\bar{Z}$ to 2J6-c
A3 and B3	T9 + 1.0 ± 0.5		32 vdc max, min of 24V average during steady stage	2J1-r to 2J1-$\bar{k}$ 2J1-e to 2J1-$\bar{k}$
TDDFL	T8 + 15 ± 5		No change	2J7-D to 2J3-h
Fire pulse	T11 + 5.4 ± 0.7		No change	

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3.11 Environmental testing. The control unit shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The control unit shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.6. Tests shall be conducted in the order specified.

3.11.1 High temperature. The control unit shall withstand a temperature of 160 + 5°F. without damage and shall meet the requirements of 3.7 and 3.8, while at this temperature when tested as specified in 4.4.6.1.

3.11.2 Low temperature. The control unit shall withstand a temperature of -65° + 5°F. without damage and shall meet the requirements of 3.7 and 3.8, while at this temperature when tested as specified in 4.4.6.2.

3.11.3 Transportation vibration. The control unit shall withstand vibration for 15 minutes in each of the three mutually perpendicular axes, as shown in Figure 5, without evidence of damage effecting function, at the frequencies and amplitudes specified in Table II when tested as specified in 4.4.6.3. The frequency shall be varied uniformly from 8 to 500 to 8 Hertz (Hz) in 15 minutes. Following this test the control unit shall meet the requirements of 3.6 through 3.9.

Table II

Frequency range (Hz)	Cycle time	Constant acceleration or double amplitude
8-55-8	7.5	0.060 ± 0.006 inch
55-500-55	7.5	10 ± 1 g

3.11.3.1 Low level transportation vibration. The control unit shall withstand vibration for 10 minutes in each of the three mutually perpendicular axes, as shown in Figure 5, without evidence of damage effecting function, at the frequencies and amplitudes specified in Table III when tested as specified in 4.4.6.3.1. The frequencies shall be varied uniformly from 8 to 500 to 8 Hz in 10 minutes. Following this test, the control unit shall meet the requirements of 3.6, 3.7 and 3.8.

Table III

Frequency range (Hz)	Cycle time	Constant acceleration or double amplitude
8-40-8	5.0 min	0.060 ± 0.006
40-500-40	5.0 min	5.0 ± 0.05 g

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3.11.4 Operational vibration. The control unit shall withstand a programed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration without evidence of damage or affecting functioning when tested as specified in 4.4.6.4. The total time of test shall be 210 seconds; 70 seconds in each of three mutually perpendicular axes, as shown in Figure 5. The complex vibration test consists of the following five sections programed continuously:

a. White Gaussian noise at a level of 22 g Root Mean Square (rms) band-limited between 15 and 1500 Hz for one (1) second.
b. A 4 g rms swept sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with frequency, one such sweep taking the full 15 seconds.
c. An 8 g rms swept sine wave superimposed on 15 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.
d. A 4 g rms sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.
e. White Gaussian noise at a level of 22 g rms band-limited between 15 and 1500 Hz for one second. The control unit shall meet the requirements of 3.7 concurrent with third axis vibration starting at X + 130 + 5 seconds of Table I.

3.11.5 Shock. The control unit shall withstand shocks of 30 + 5 g's three times in each direction along the three mutually perpendicular axes, shown in Figure 5, for a total of 18 shocks when tested as specified in 4.4.6.5. Following this test, the assembly shall meet the requirements of 3.6 through 3.9.

3.12 Test codes. The contractor shall provide data feed back on punch cards in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified within the paragraphs in section 3.

3.13 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

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3.14 Workmanship.

3.14.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.14.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.14.3 Burrs. Parts shall be free from burrs.

3.14.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.14.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.14.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8830142) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV. Tests shall be conducted in order specified in Table IV.

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Table IV

First article inspections and tests

Inspection or Test	Dest	Non-Dest	Testing Sequence for Unit Nos. 1	2	3
1. Classification of characteristics (see 4.3.2.1)		X			
2. Continuity and resistance (see 3.6)		X	1	1	1
3. Insulation resistance (see 3.8)		X	2	2	2
4. Diode characteristics (see 3.7)		X	3	3	3
5. Pressure retention (see 3.9)		X	4	4	4
6. Transportation vibration (see 3.11.3)		X	5	5	
7. Operation vibration (see 3.11.4)	X				5
8. Shock (see 3.11.5)	X		6	6	
9. High temperature (see 3.11.1)		X	7		
10. Low temperature (see 3.11.2)		X		7	
11. Operation (see 3.10)		x	8*	8**	

*This test to be conducted as the final test for Unit No. 1 while the control unit is at high temperature.

**This test to be conducted as the final test for Unit No. 2 while the control unit is at low temperature.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing

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the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table V and Table VI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Adaption kit sub-assembly, control unit,M4 (dwg 8830142).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Thermal release device missing	Visual
Major:		
	101. Location of mounting holes	Gage
	102. Diameter of mounting holes (max)	Gage
	103. Objective evidence relative to material missing or improper (see 3.1)	Visual
	104. Serialization improper or missing (see 3.4)	Visual
Minor:		
	201. Improper torque of screws	Gage
	202. Component loose	Visual/Manual
	203. Component missing	Visual
	204. Incorrect assembly	Visual
	205. Component damaged	Visual
	206. Incorrect or illegible marking	Visual
	207. Bare spot in protective coating except for slight scratches	Visual
	208. Evidence of poor workmanship (see 3.14)	Visual

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4.3.2.1.2 Cover, control unit (dwg 8829845).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper location of 12 mounting holes	Gage
102.	Hole size out of tolerance (12 holes)	Gage
103.	Improper loaction and alignment of 5 slots	Gage
104.	Improper width of slots (5)	Gage
Minor:		
201.	Defects in "O" ring groove	Visual

4.3.2.1.3 Control housing assembly, prior to installation of inserts (dwg 8838334).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of 14 inner threaded holes (on mounting bases)	Gage
102.	Pitch diameter of 14 inner threaded holes, max	Gage
103.	Minor diameter of 14 inner threaded holes, max	Gage
104.	Location of 12 threaded holes on flange	Gage
105.	Pitch diameter of 12 threaded holes on flange, max	Gage
106.	Minor diameter of 12 threaded holes on flange, min	Gage
107.	Location of 18 threaded holes on flat side	Gage
108.	Pitch diameter of 18 threaded holes on flat side, max	Gage
109.	Minor diameter of 18 threaded holes on flat side, max	Gage
Minor:		
201.	Height of mounting bases	Gage
202.	Total height of housing	Gage

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4.3.2.1.4 Control housing assembly after installation of inserts (see dwg 8838334).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorect assembly of inserts	Visual/Gage
	102. Cracked inserts	Visual
	103. Loose inserts	Visual/Manual
	104. Missing inserts	Visual
Minor:	None defined.	

4.3.2.1.5 Preparation for delivery (see dwg 8830142).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packaging and packing not in accordance with dwg.	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Workmanship (see 3.14)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Continuity and resistance	Major
b. Circuit resistance	Major
c. Diode characteristics	Major
d. Insulation resistance	Major
e. Low level transportation vibration	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity and resistance, diodes characteristics, insulation resistance, and pressure retention. Each control unit shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.6 through 3.9, the item shall be rejected and removed from the lot.

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4.3.3.2 Low level tansportation vibration. A sample shall be selected at random in accordance with Table VII . The sample shall be tested in accordance with 4.4.6.3.1. If any item fails to comply with 3.11.3.1, the lot shall be rejected.

Table VII

Lot Size	Sample Size
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7
1301-3200	10

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8854157). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity and resistance. The continuity and resistance shall be measured between the terminals listed in 3.6. To energize relays, apply 115 ± 10 vdc between terminals 2J4-W and 2J4-d.

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4.4.1.1 Circuit resistance. The circuit resistance between terminals listed in 3.6.1 shall be measured with a four wire resistance measuring instrument, except for inductive resistances or resistances greater than 1000 ohms which will be measured with a digital ohmmeter of sufficient accuracy or a wheatstone bridge. Current through the squib circuits shall be limited to 5 milliamps.

4.4.2 Diode characteristics. Diodes shall be tested for forward voltage drop when the current specified in 3.7 is applied in the forward direction. Reverse current shall be determined by the voltage drop across a series resistor of 150 ± 1.5 ohms when voltage is applied in the reverse direction. The duration of the tests in either the forward or reverse direction shall not be more than 15 seconds.

4.4.3 Insulation resistance. The insulation resistance of the control unit shall be measured between the terminals listed in 3.8 using a test voltage of 500 ± 50 volts with current limited to 5 milliamperes.

4.4.4 Pressure retention. The control unit shall be pressurized with Nitrogen Gas, Type I, Class I, Grade A or B BB-N-411. The pressure source shall be shut off and the pressure observed for a period of one hour minimum.

4.4.5 Operation. The control unit shall be connected to the simulator circuits shown in Figures 1 through 4. The power supplies in the Parameter Insertion Simulator shall be energized, and the switches shall be closed as specified in Table I. An oscillogram shall be made of the voltage signals fed into the warhead simulator terminals specified in 3.10.

4.4.6 Environmental testing (see 3.11).

4.4.6.1 High temperature. The control unit shall be placed in a test chamber. The chamber's temperature shall be raised and stabilized at 160 ± 5°F (see 6.4). After the control unit has been subjected to this stabilized temperature for a mimimum of 4 hours and while still at this temperature, it shall be tested for compliance with requirements of 3.7 and 3.8.

4.4.6.2 Low temperature. The control unit shall be placed in a test chamber. The chamber's temperature shall be lowered and stabilized at -65 ± 5°F (see 6.4). After the control unit has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.7 and 3.8.

4.4.6.3 Transportation vibration. The control unit shall be suitably mounted on a vibration machine and vibrated for 15 minutes in each of three mutually perpendicular axes, as shown in Figure 5, at the frequencies and amplitudes specified in 3.11.3. Total test time shall be 45 minutes. After the completion of the test, the control unit shall be tested for compliance with 3.6 through 3.9.

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4.4.6.3.1 Low level transportation vibration. The control unit shall be suitably mounted on a vibration machine and vibrated for 10 minutes in each of three mutually perpendicular axes, as shown in Figure 5, at the frequencies and amplitudes specified in 3.11.3.1. Total test time shall be 30 minutes. Upon completion of the vibrations the control unit shall be tested for compliance with 3.6, 3.7 and 3.8. After successful completion of the post-vibration electrical tests, the control unit shall be inspected, externally and internally, for improper torque of screws, loose components, and damaged components.

4.4.6.4 Operational vibration. The control unit shall be mounted on a vibration machine and vibrated in three mutually perpendicular axes, as shown in Figure 5, using a standard noise or complex wave tape. The tape shall have the characteristics listed in the following table.

Time (sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (calibration) (1)	1.	Set gain to give 2 g rms
20-30	None	0	0
30-31	Noise (1)	11.	22.
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
46-61	Noise (1) plus sinusoid (3)	8.5	17.
	Noise alone	7.5	15.
	Sinusoid alone	4.	8.
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
99-100	Noise (1)	11.	22.

(1) Band-limited (15-1500 Hz) white Gaussian Noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

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For creating the random noise, the output of a General Radio random noise generator, or its equivalents, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring correct noise bandwidth, a filter with an asymptotic slope of at least 24 Decibel (Db) per octave and 3 Db points at 15 and 1500 Hz shall be considered acceptable. Prior to the start of vibration in the third axis, the control unit shall be activated as specified in 3.10. At X + 130 ± 5 seconds vibration in the third axis shall begin. Inspection shall be made for compliance with requirements of 3.10.

4.4.6.4.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire vibration system shall be as flat as is consistent with the present state of the art. Plus or minus 3 Db or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system the standard noise tape is applied. The over-all gain of the shake system is then adjusted so that with the noise calibration signal, the g level listed in 4.4.6.4 is obtained. With the same gain setting, the tape is run for the remainder of the test and the g levels obtained will conform to the levels listed in the table of 4.4.6.4.

4.4.6.5 Shock. The control unit shall be rigidly mounted to a test fixture which shall be mounted to suitable shock producing apparatus. The control unit shall then be subjected to shocks as specified in 3.11.5. Duration of shock shall be 11 ± 2 millisecond as measured between points which are 10% of peaks. The time to peak shall be 5.5 ± 1 millisecond. After the shocks the control unit shall be examined for evidence of damage and tested for compliance with requirements of 3.6 through 3.9.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The control unit shall be packaged in accordance with dwg 8830142 using the specified moisture barrier.

5.1.2 Level C. The control unit shall be packaged in accordance with dwg 8830142 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The control unit shall be packed in accordance with dwg 8830142.

5.3 Marking. Marking shall be as specified in accordance with dwg 8830142.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 3.4, 3.12 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2068 (Rev.1) dated 6 May 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A156

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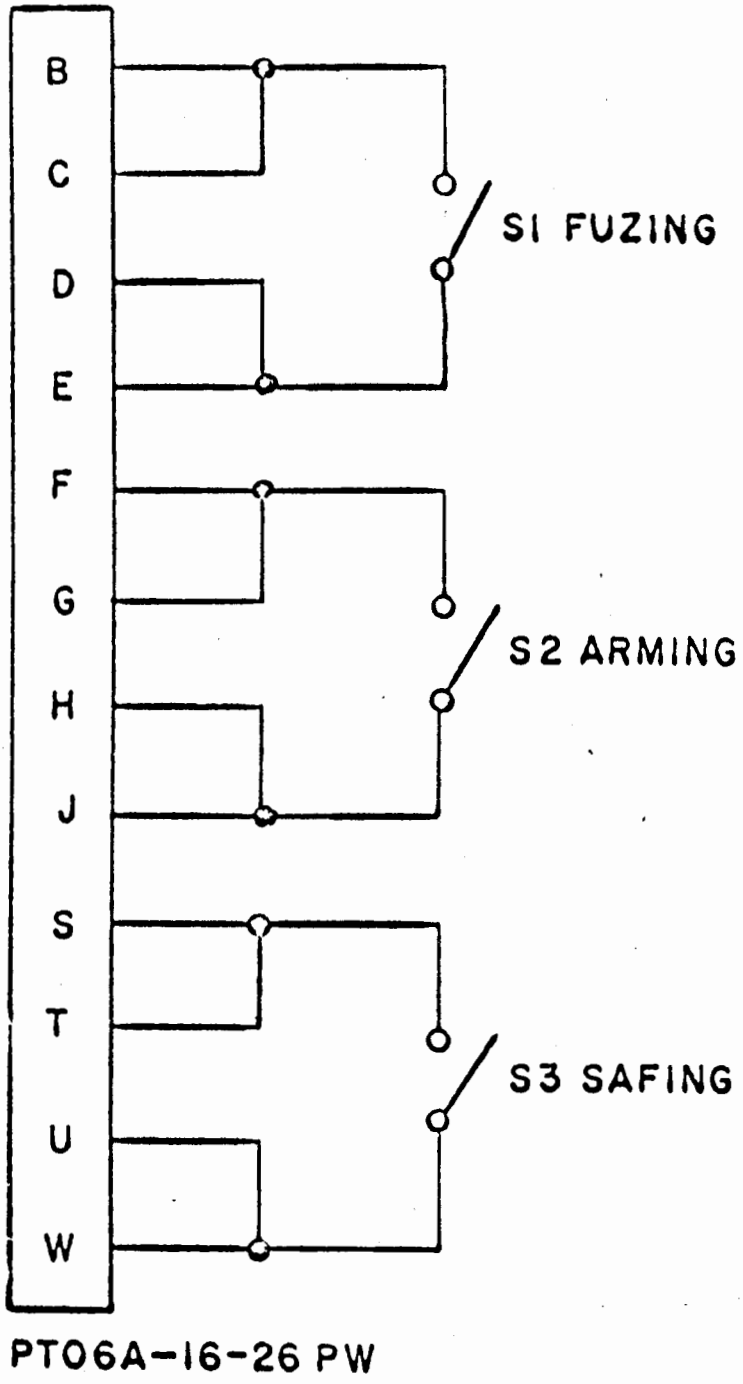


Figure 1 - Baro switch simulator

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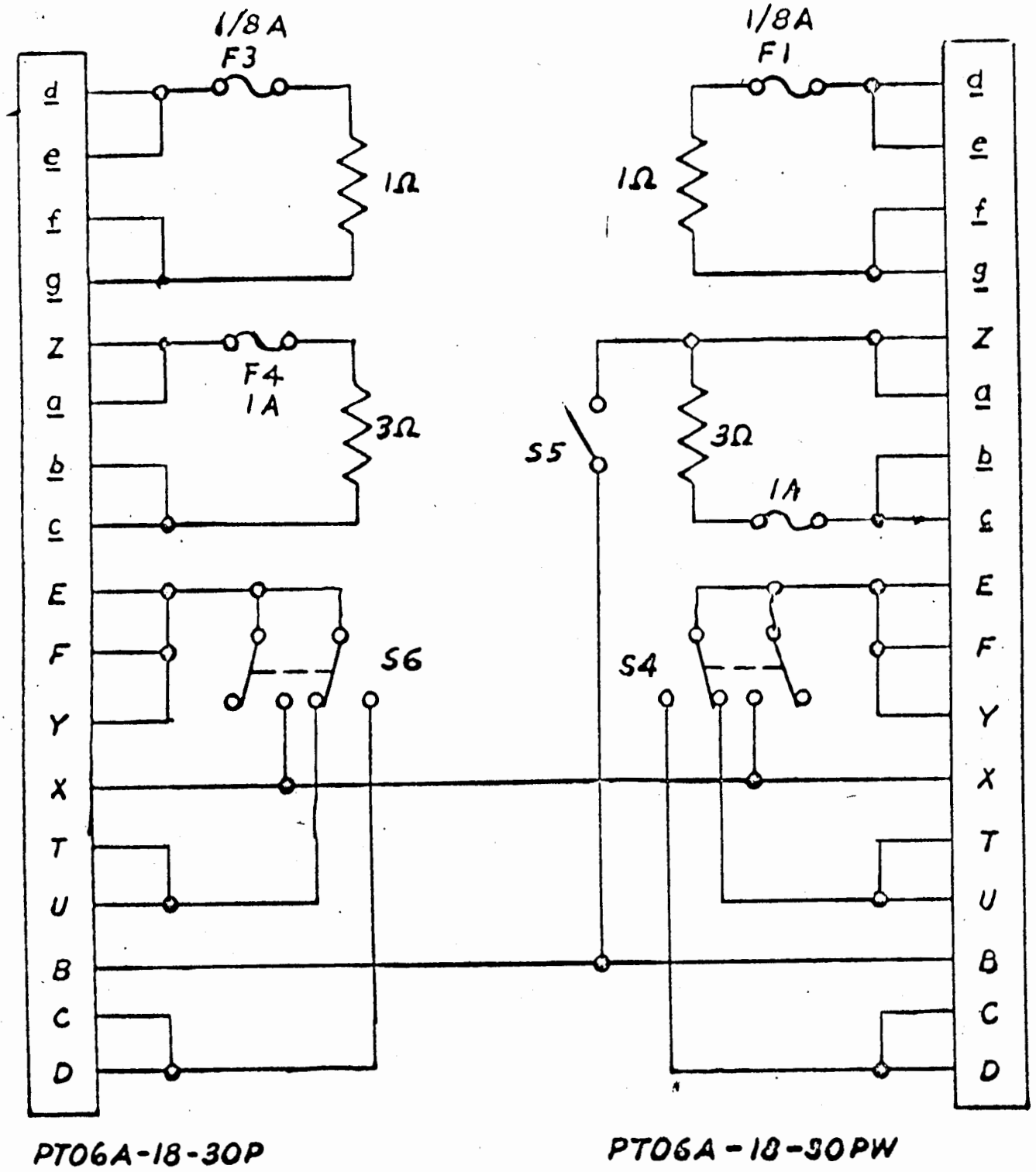


Figure 2 - TDD simulator

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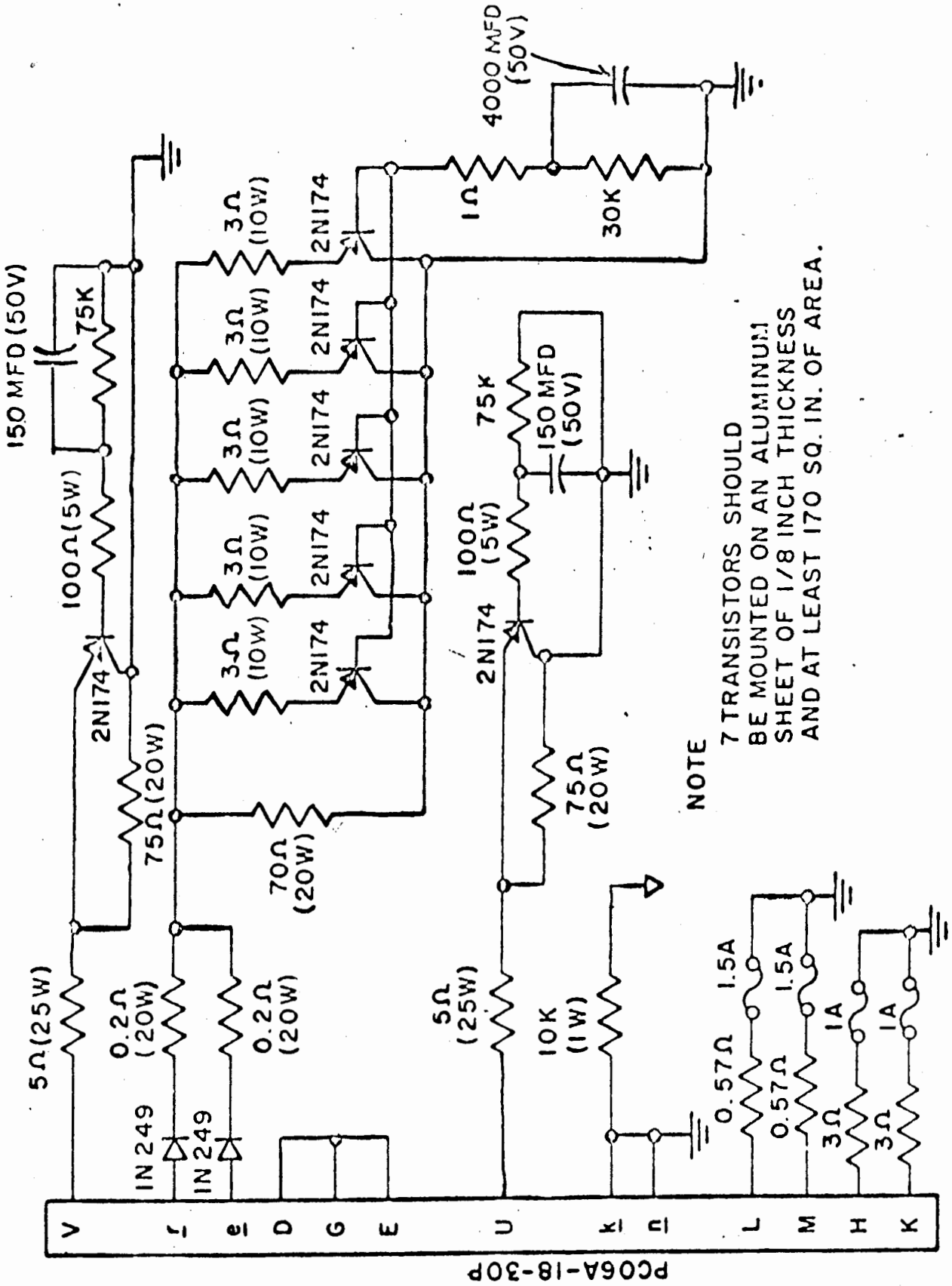


Figure 3 – Warhead simulator

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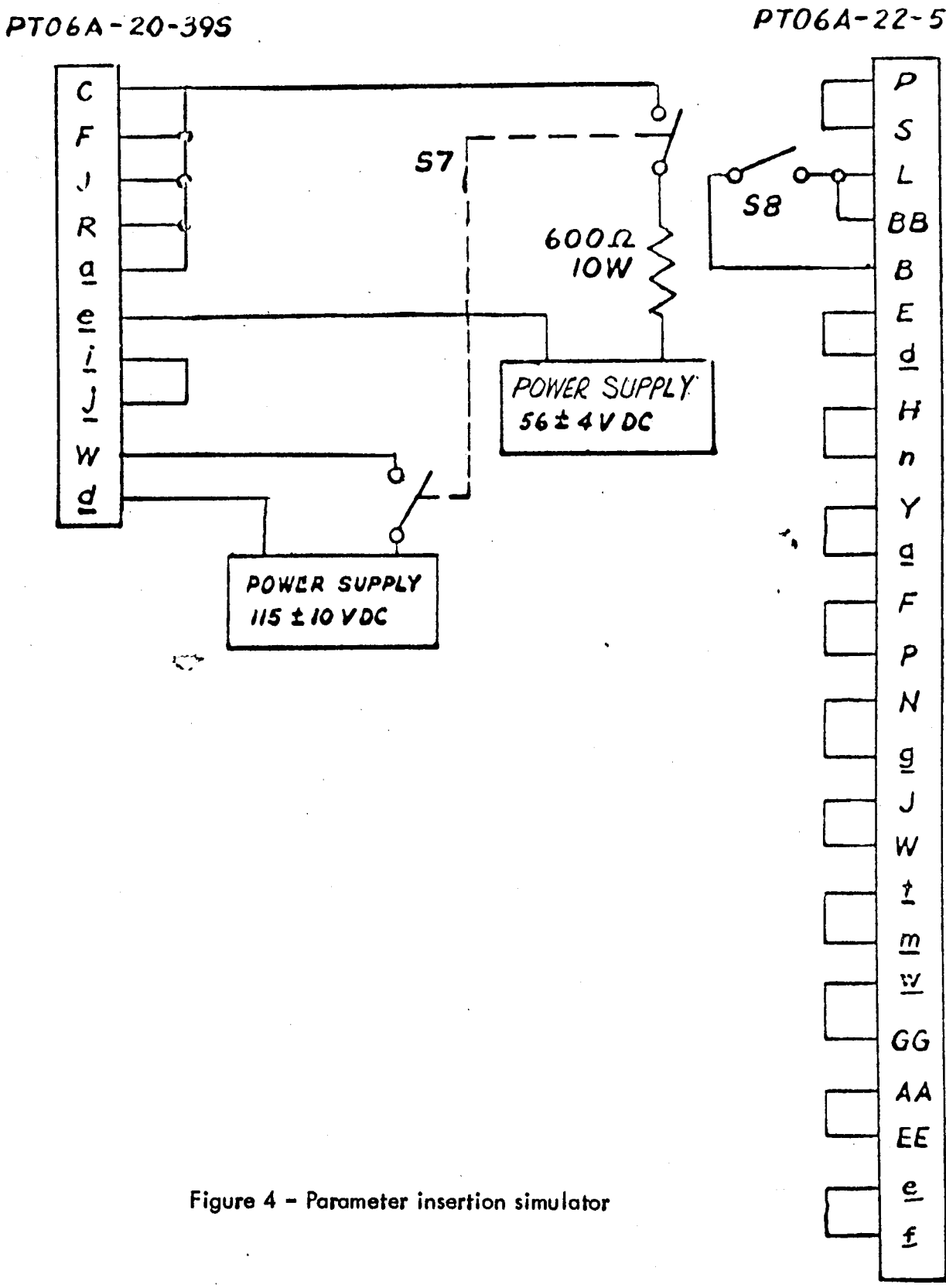


Figure 4 - Parameter insertion simulator

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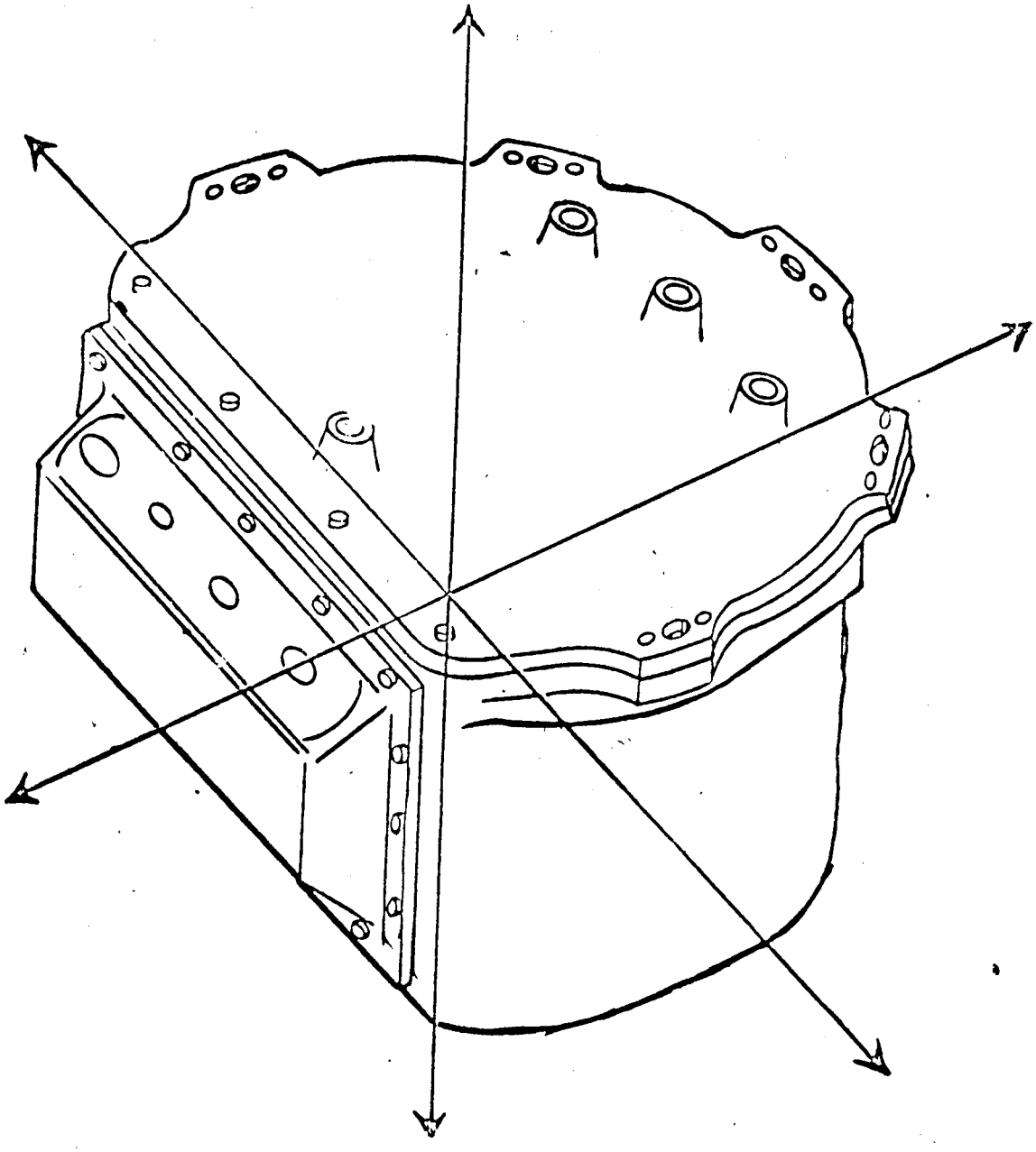


Figure 5 - Axes for vibration and shock of the control unit

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